Part 1: Preprocessing

Filter-bank CSP + band power in short time windows

ML Frontiers, VISTEC · BCIC IV-2a, left vs right hand MI

Why not just use CSP, or MixNet's input?

- MI shows up as ERD/ERS in Mu and beta, around C3/C4 [lecture p.51]
- Classic CSP keeps one log-variance per trial, so it throws away the timing inside the trial [lecture p.62]
- MixNet keeps the whole CSP time series [lecture p.63]: 7,200 numbers per trial, learned from only ∼115 training trials

	Classic FBCSP	MixNet	Ours
Keeps timing	no	all 400 samples	15 windows
Shape per trial	36	18×400	36×15
Numbers per trial	36	7,200	540

Hypothesis

Band power in short time windows keeps the timing of ERD/ERS, at a size a small model can learn from ∼115 trials.

Pipeline: four steps, two of them learned



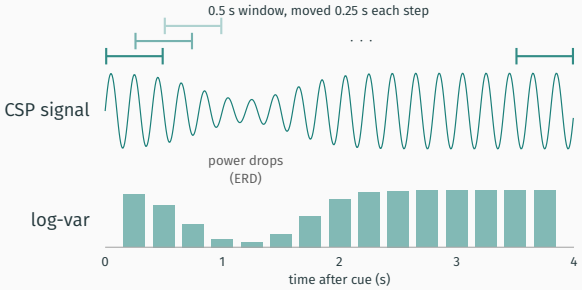
Teal boxes: fit on the training fold only

- "Fit CSP: training data only" [lecture p.60]
- "Training-set per-channel standardization" [lecture p.31]
- Validation and session E only reuse the saved filters and mean/SD. A self-check in the code tests this.

Output per trial

36 features (9 bands × 4 CSP filters) × 15 time windows, standardized

Our change: one log-variance per short window



Normalized log-variance

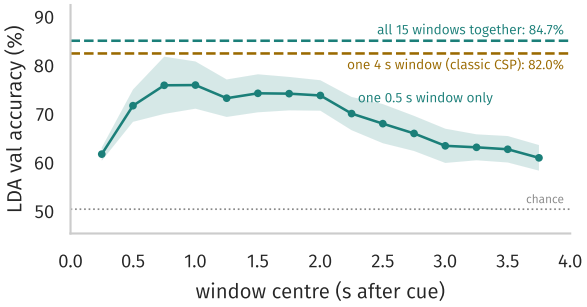
For band b, CSP filter i, window w:

$$f_{b,i,w} = \log \frac{\operatorname{var}(z_{b,i}[w])}{\sum_{j=1}^{4} \operatorname{var}(z_{b,j}[w])}$$

[lecture p.59]

- $(400 - 50)/25 + 1 = 15$ windows in 4 s
- This step learns nothing, so it cannot leak
- Then each feature is standardized with the training-fold mean and SD [lecture p.31]

Quick LDA test: class information is not spread evenly in time

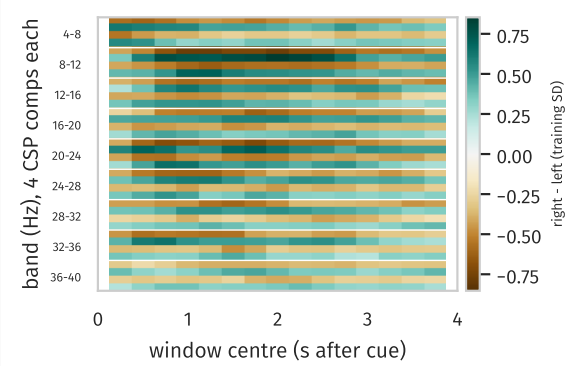


- One 0.5 s window (36 numbers, one LDA per window): peaks at 0.75-1 s (75.5%), near 61% at both ends
- One 4 s window = whole trial, classic CSP (36 numbers): 82.0%
- All 15 windows together = our features (540 numbers): 84.7%, better in 8 of 9 subjects

How we got these numbers

A simple shrinkage LDA, the classic classifier after CSP [lecture p.60], trained on our features before building HWNet. Same 5 folds as the real pipeline: fit on the training fold, score on the validation fold of session T, mean of 9 subjects (band = ±1 SE). Session E not used: not test results, not comparable with MixNet.

What the model sees: 36 features × 15 windows



All 144 session T trials per subject, averaged over 9 subjects. Session E not used.

- Each cell: right minus left class mean, in training SD units
- Strongest in 8-12 Hz (Mu) and 20-24 Hz (beta), from about 0.5 to 2.5 s, the bands where ERD/ERS shows up [lecture p.51]
- Rows alternate in sign because CSP takes filters from both ends: one end has more power for left hand, the other for right [lecture p.57]

Takeaway

Each trial becomes *which band*, *which spatial filter* and *when*, in 540 numbers.

Backup: settings and data notes

All settings live in `experiments/prep_HW.py`

Setting	Ours	Ablation
`win_len`, `win_step`	0.5 s, 0.25 s	4.0 s, 4.0 s (classic)
`n_components`	4	2 (MixNet size)
`norm_logvar`	True	False
`bands`	9 bands, 4-40 Hz	
`order`	5	

Quick LDA check (validation, session T): ours 84.7%, classic 82.0%, `norm_logvar=False` 85.0%, `n_components=2` not run yet.

Data notes

- 50 Hz line noise: removed. After resampling to 100 Hz it sits at the top edge, where every band filter blocks it
- Drift and eye movement below 4 Hz: removed by the lowest band edge
- Not removed: blinks in 4-8 Hz, muscle noise in beta, and trials marked as artifacts (2 to 36 of 144 per subject)